Bijectivity and the double angle formula

November 8, 2026

Injection, surjection, and bijection

Let us repeat the definitions from class. Suppose $f : X \mapsto Y$ is a map from X to Y.

Def: f is **injective** if

$$\forall\ a, b \in X,\ a \neq b \implies f(a) \neq f(b)$$

This is also called a **one-to-one** map.

Def: f is **surjective** if

$$\forall\ y\ \in\ Y\ \exists\ x\ \in\ X \text{ such that } f(x) = y$$

This is also called an **onto** map.

Def: f is **bijective** if it is bother injective and surjective

1. Draw a picture to illustrate a map which is injective but not surjective
2. Draw a map which is surjective but not injective
3. Prove that a function would have to be bijective in order to be invertible. Proof outline: Consider what happens to the domain and range of a function (including all points within them) when it is inverted. Now consider why invertibility will fail if a function is not injective, and then again if it is not surjective.

Double angle formula

The sine double angle formula is

$$sin(2\theta) = 2sin(\theta)cos(\theta)$$

Draw a unit circle and inscribe a triangle such that two points are on the circle, and one point is at the centre of the circle. An illustration is included below. Use the **Inscribed Angle Theorem** to prove the double angle formula above as follows.

1. Come up with an expression for the area of the triangle (You might choose to use the area of the parallelogram formed by the triangle)
2. Bisect that triangle into two right triangles. Find an expression for the area for these right triangles.
3. Note that the area of the large triangle must be the same as the area of the sum of the two small triangles.

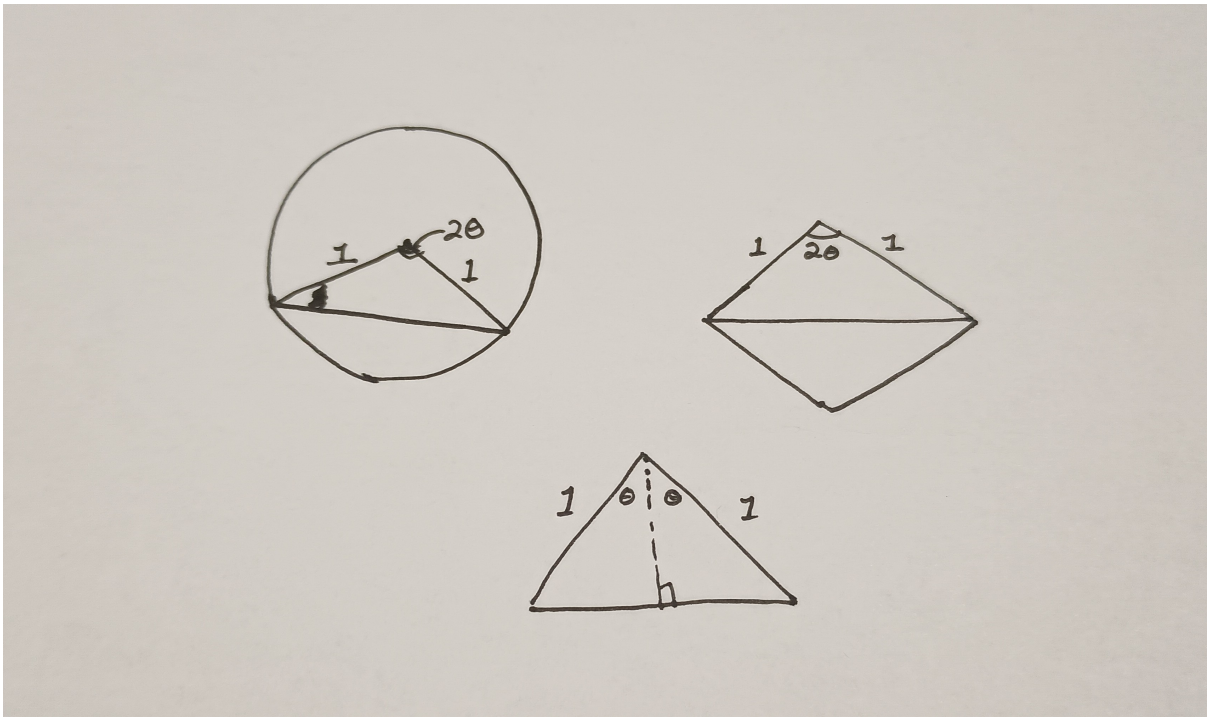


Figure 1: Geometry of proof